

Gaussian fluctuations and breakdown of Mean-Field Theory

Luca Dall'Asta

DISAT
Politecnico di Torino, Italy

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Validity of the Landau approximation

The Landau approximation neglects fluctuations around the saddle point (i.e. fluctuations of $m(\vec{x})$ around the uniform mean-field value M_0).

Ginzburg criterion:

$$\langle (\Delta m)^2 \rangle \ll M_0^2 \quad (1)$$

The Landau MF theory should be incorrect when this condition is violated (close to criticality):

- r.h.s.: $M_0 \sim |T - T_0|^{1/2}$,
- l.h.s.: $\langle (\Delta m)^2 \rangle \sim |T - T_0|^{??}$ (let's compute it...)

Validity of the Landau approximation

$$\langle (\Delta m)^2 \rangle = \xi^{-2d} \int_{B(\xi)} d\vec{x} d\vec{y} \langle (m(\vec{x}) - M_0) (m(\vec{y}) - M_0) \rangle \quad (2)$$

$$= \xi^{-d} \int_{B(\xi)} d\vec{r} G(\vec{r}) \quad (3)$$

$$= \xi^{-d} \int_0^\xi dr r^{d-1} r^{2-d} Y\left(\frac{r}{\xi}\right) \quad (4)$$

$$\propto \xi^{2-d} \int_0^1 dz z Y(z) \sim |T - T_0|^{\frac{d-2}{2}}, \quad (5)$$

where, $G(r) = r^{2-d} Y(r/\xi) \propto c_0^{-1} r^{2-d}$ and $\xi \sim |T - T_0|^{-1/2}$.

The Ginzburg criterion can be written as

$$|T - T_0|^{\frac{d-2}{2}} \ll |T - T_0| \iff d > 4, \quad (6)$$

i.e. for $d > 4$, the ferro-para transition is of mean-field type.

Ginzburg criterion: general formulation

In a general mean-field theory, critical behavior is given by

$$M_0 \sim |T - T_0|^{\beta_{MF}}, \quad \xi \propto |T - T_0|^{-\nu_{MF}}, \quad \chi_T \propto |T - T_0|^{-\gamma_{MF}}, \quad (7)$$

the Ginzburg criterion becomes

$$|T - T_0|^{-\gamma_{MF} + d\nu_{MF}} \ll |T - T_0|^{2\beta_{MF}} \quad (8)$$

or equivalently

$$d > d_{uc} = \frac{2\beta_{MF} + \gamma_{MF}}{\nu_{MF}}. \quad (9)$$

REMARK: For ferro-para transition $\beta_{MF} = 1/2$, $\gamma_{MF} = 1$, $\nu_{MF} = 1/2$, and we retrieve $d_{uc} = 4$.

Ginzburg criterion: general formulation

REMARK: Landau's MF theory is again valid well below the critical point ($T \leq T_G < T_0$).

Interaction range:

$$R^2 = \frac{J_2}{J_0} = \frac{\int d\vec{r} r^2 J(r)}{\int d\vec{r} J(r)} \propto \frac{c_0}{r_0}. \quad (10)$$

Since

$$M_0^2 = -\frac{r_0 t}{u_0} \sim r_0 |t| \sim \frac{c_0}{\xi^2}, \quad \langle (\Delta m)^2 \rangle \sim \frac{\xi^{2-d}}{c_0} \quad (11)$$

where $\xi \propto \sqrt{c_0/(r_0 t)} = R t^{-1/2}$, then

$$\xi^{4-d} \ll c_0^2 \propto R^4 \quad \implies \quad |t| \gg t_G \simeq R^{2d/(d-4)}. \quad (12)$$

For $d < 4$, if R is large, the condition can be satisfied pretty close to criticality (e.g. superconductivity).

Corrections to the saddle point

We can expand $\mathcal{H}_{\text{eff}}$ at the second order around the saddle-point solutions

$$M_0 = \begin{cases} 0 & T > T_0 \\ \pm \sqrt{-r_0 t / u_0} & T < T_0 \end{cases} \quad (13)$$

introducing the fluctuation field $\delta m(\vec{x}) = m(\vec{x}) - M_0$,

$$\begin{aligned} \beta \mathcal{H}_{\text{eff}}[\delta m] &= \int d\vec{x} \left[\frac{c_0}{2} (\nabla (M_0 + \delta m(\vec{x})))^2 + \frac{r_0 t}{2} (M_0 + \delta m(\vec{x}))^2 + \frac{u_0}{4} (M_0 + \delta m(\vec{x}))^4 \right] \\ &= \int d\vec{x} \left[\frac{c_0}{2} (\nabla \delta m)^2 + \frac{r_0 t}{2} M_0^2 + r_0 t M_0 \delta m(\vec{x}) + \frac{r_0 t}{2} \delta m(\vec{x})^2 \right. \\ &\quad \left. + \frac{u_0}{4} M_0^4 + u_0 M_0^3 \delta m(\vec{x}) + \frac{3}{2} u_0 M_0^2 \delta m(\vec{x})^2 + u_0 M_0 \delta m(\vec{x})^3 + \frac{u_0}{4} \delta m(\vec{x})^4 \right] \\ &= \int d\vec{x} \left[\frac{c_0}{2} (\nabla \delta m)^2 + (r_0 t M_0 + u_0 M_0^3) \delta m(\vec{x}) \right. \\ &\quad \left. + \left(\frac{r_0 t}{2} + \frac{3}{2} u_0 M_0^2 \right) \delta m(\vec{x})^2 + O(\delta m^3) + \text{const.} \right] \end{aligned}$$

Corrections to the saddle point

By noticing that $r_0 t M_0 + u_0 M_0^3 = 0$ and setting

$$\rho_0(t) = r_0 t + 3u_0 M_0^2 = \begin{cases} r_0 t & T > T_0 \\ -2r_0 t & T < T_0 \end{cases} \quad (14)$$

we find

$$\beta \mathcal{H}_{\text{eff}}[\delta m] \approx \int d\vec{x} \left[\frac{c_0}{2} (\nabla \delta m)^2 + \frac{\rho_0(t)}{2} \delta m(\vec{x})^2 \right]. \quad (15)$$

Rescaling field $\delta m(\vec{x}) = a^{d/2} \phi(\vec{x})$ and parameters

$$c = a^d c_0 = \beta \bar{c}, \quad \rho(t) = a^d \rho_0(t) = \beta \bar{\rho}(t) \quad (16)$$

we obtain the effective Hamiltonian of the **Gaussian model**,

$$\mathcal{H}_G[\phi] = \int d\vec{x} \left[\frac{\bar{c}}{2} (\nabla \phi)^2 + \frac{\bar{\rho}(t)}{2} \phi(\vec{x})^2 \right] \quad (17)$$

Gaussian Model: exact calculations

The Fourier representation of the partition function reads,

$$Z_G[0] = \int \frac{\prod_{\vec{k}: k_x > 0}^{|\vec{k}| < \Lambda} d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k})}{\mathcal{N}(V, \Lambda)} \exp \left[-\frac{\beta}{2V} \sum_{\vec{k}} (\bar{c}k^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k})|^2 \right] \quad (18)$$

$$= \frac{1}{\mathcal{N}(V, \Lambda)} \prod_{\vec{k}: k_x > 0}^{|\vec{k}| < \Lambda} \left\{ \int d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k}) \exp \left[-\frac{\beta}{V} (\bar{c}k^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k})|^2 \right] \right\} \quad (19)$$

$$= \mathcal{N}(V, \Lambda)^{-1} \prod_{\vec{k}: k_x > 0}^{|\vec{k}| < \Lambda} \frac{2\pi V k_B T}{\bar{c}k^2 + \bar{\rho}(t)} \quad (20)$$

$$= \mathcal{N}(V, \Lambda)^{-1} \prod_{\vec{k}}^{|\vec{k}| < \Lambda} \left(\frac{2\pi V k_B T}{\bar{c}k^2 + \bar{\rho}(t)} \right)^{\frac{1}{2}}. \quad (21)$$

REMARK: We integrated over complex variables, using

$$\int \frac{dz dz^*}{2i} e^{-az^* z} = \int dx dy e^{-a(x^2 + y^2)} = \frac{\pi}{a}, \quad (22)$$

Gaussian Model: exact calculations

Adding an external field $h(\vec{x})$ and exploiting properties of functional gaussian integrals,

$$Z_G[h] = \int \frac{\prod_{\vec{k}: k_x > 0}^{|k| < \Lambda} d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k})}{\mathcal{N}(V, \Lambda)} \exp \left[-\frac{\beta}{2V} \sum_{\vec{k}} \left(\bar{c} k^2 + \bar{\rho}(t) \right) |\hat{\phi}(\vec{k})|^2 + \frac{\beta}{V} \sum_{\vec{k}} \hat{h}(\vec{k}) \hat{\phi}(-\vec{k}) \right] \quad (23)$$

$$\propto \exp \left[\frac{1}{2} \sum_{\vec{k}}^{|k| < \Lambda} \log \frac{2\pi V k_B T}{\bar{c} k^2 + \bar{\rho}(t)} + \frac{\beta}{2V} \sum_{\vec{k}} \frac{\hat{h}(\vec{k}) \hat{h}(-\vec{k})}{\bar{c} k^2 + \bar{\rho}(t)} \right] \quad (24)$$

$$= Z_G[0] e^{\frac{\beta}{2V} \sum_{\vec{k}} \frac{\hat{h}(\vec{k}) \hat{h}(-\vec{k})}{\bar{c} k^2 + \bar{\rho}(t)}}. \quad (25)$$

Gaussian Model: exact calculations

The free-energy is

$$F[h] = -\frac{1}{\beta} \log Z_G[h] \quad (26)$$

$$= -\frac{k_B T}{2} \sum_{\vec{k}} \log \left(\frac{2\pi V k_B T}{\bar{c} k^2 + \bar{\rho}(t)} \right) - \frac{1}{2V} \sum_{\vec{k}} \frac{|\hat{h}(\vec{k})|^2}{\bar{c} k^2 + \bar{\rho}(t)} \quad (27)$$

REMARK: The first term determines a divergence in the $V \rightarrow \infty$ limit (even for free energy density $f = F/V$). This is consistent with the existence of an infinite number of degrees of freedom. Physical quantities are not affected and the diverging free energy can be fixed by appropriately choosing the normalization factor $\mathcal{N}(V, \Lambda)$.

Gaussian Model: exact calculations

The *two-point correlation function* can be easily computed in the Fourier space ($\prod'$ stays for the constrained product),

$$\langle \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \rangle = \frac{\int \left(\prod_{\vec{k}}' d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k}) \right) \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) e^{-\frac{\beta}{2V} \sum_{\vec{k}} (\bar{c}k^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k})|^2}}{\int \left(\prod_{\vec{k}}' d\hat{\phi}(\vec{k}) d\hat{\phi}^*(\vec{k}) \right) e^{-\frac{\beta}{2V} \sum_{\vec{k}} (\bar{c}k^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k})|^2}} \quad (28)$$

$$= \frac{\int \prod_{i=1,2} \left\{ d\hat{\phi}(\vec{k}_i) d\hat{\phi}^*(\vec{k}_i) e^{-\frac{\beta}{V} (\bar{c}k_i^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k}_i)|^2} \right\} \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2)}{\prod_{i=1,2} \left\{ \int d\hat{\phi}(\vec{k}_i) d\hat{\phi}^*(\vec{k}_i) e^{-\frac{\beta}{V} (\bar{c}k_i^2 + \bar{\rho}(t)) |\hat{\phi}(\vec{k}_i)|^2} \right\}}. \quad (29)$$

Three possible cases:

- if $\vec{k}_1 \neq \pm \vec{k}_2$, the calculation factorizes, so that

$$\langle \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \rangle = \langle \hat{\phi}(\vec{k}_1) \rangle \langle \hat{\phi}(\vec{k}_2) \rangle = 0 \quad (30)$$

Gaussian Model: exact calculations

- if $\vec{k}_1 = \vec{k}_2$, using polar coordinates $\hat{\phi}(\vec{k}_1) = \left| \hat{\phi}(\vec{k}_1) \right| e^{i\theta} = r e^{i\theta}$,

$$\left\langle \hat{\phi}(\vec{k})^2 \right\rangle = \frac{\int_0^\infty r dr \int_0^{2\pi} d\theta e^{2i\theta} r^2 e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))r^2}}{\int_0^\infty r dr \int_0^{2\pi} d\theta e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))r^2}} \quad (31)$$

$$\propto \int_0^{2\pi} d\theta e^{2i\theta} = 0 \quad (32)$$

- if $\vec{k}_1 = -\vec{k}_2$,

$$\left\langle \left| \hat{\phi}(\vec{k}) \right|^2 \right\rangle = \frac{\int_0^\infty r dr \int_0^{2\pi} d\theta r^2 e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))r^2}}{\int_0^\infty r dr \int_0^{2\pi} d\theta e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))r^2}} \quad (33)$$

$$= \frac{\int_0^\infty dw w e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))w}}{\int_0^\infty dw e^{-\frac{\beta}{V}(\bar{c}k^2 + \bar{\rho}(t))w}} \quad (34)$$

$$= \frac{k_B T V}{\bar{c}k^2 + \bar{\rho}(t)}. \quad (35)$$

Gaussian Model: exact calculations

The real-space two-point correlation function reads

$$G(\vec{x} - \vec{x}') = \langle \phi(\vec{x}) \phi(\vec{x}') \rangle \quad (36)$$

$$= \frac{1}{V^2} \sum_{\vec{k}, \vec{k}'} e^{i(\vec{k} \cdot \vec{x} + \vec{k}' \cdot \vec{x}')} \langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle \quad (37)$$

$$= \frac{1}{V^2} \sum_{\vec{k}, \vec{k}'} e^{i(\vec{k} \cdot \vec{x} + \vec{k}' \cdot \vec{x}')} \frac{k_B T V \delta_{\vec{k} + \vec{k}', 0}}{\bar{c} k^2 + \bar{\rho}(t)} \quad (38)$$

$$= \frac{1}{V} \sum_{\vec{k}} e^{i\vec{k} \cdot (\vec{x} - \vec{x}')} \frac{k_B T}{\bar{c} k^2 + \bar{\rho}(t)}. \quad (39)$$

Defining $\xi^2 = \bar{c}/\bar{\rho}(t) = c/\rho(t)$, the Fourier transform gives

$$G(r) \propto \begin{cases} r^{-\frac{d-1}{2}} e^{-\frac{r}{\xi}} & \text{for } r \gg \xi \\ r^{2-d} & \text{for } r \ll \xi \end{cases} \quad (40)$$

Gaussian Model: operator formalism

The effective Hamiltonian can be written as

$$\mathcal{H}_G[\phi] = \frac{1}{2} \int d\vec{x} d\vec{y} \phi(\vec{x}) K(\vec{x}, \vec{y}) \phi(\vec{y}) \quad (41)$$

where we defined the differential operator

$$K(\vec{x}, \vec{y}) = \left(\bar{\rho}(t) - \bar{c} \nabla_x^2 \right) \delta(\vec{x} - \vec{y}). \quad (42)$$

Performing the functional Gaussian integral, the partition function becomes

$$Z_G[h] = \int D\phi e^{-\frac{\beta}{2} \int d\vec{x} \int d\vec{y} \phi(\vec{x}) K(\vec{x}, \vec{y}) \phi(\vec{y}) + \beta \int d\vec{x} h(\vec{x}) \phi(\vec{x})} \quad (43)$$

$$= Z_G[0] \exp \left\{ \frac{\beta}{2} \int d\vec{x} \int d\vec{y} h(\vec{x}) K^{-1}(\vec{x}, \vec{y}) h(\vec{y}) \right\}, \quad (44)$$

$K^{-1}(\vec{x}, \vec{y})$ is the *Green's function* of $K(\vec{x}, \vec{y})$ or *propagator* in QFT.

Gaussian Model: operator formalism

Expectation values can be computed using the functional differentiation

$$\langle \phi(\vec{x}) \rangle = - \left. \frac{\delta F[h]}{\delta h(\vec{x})} \right|_{h=0} = \int d\vec{y} K^{-1}(\vec{x}, \vec{y}) h(\vec{y}) \Big|_{h=0} = 0, \quad (45)$$

we then find

$$G(\vec{x} - \vec{y}) = - \frac{1}{\beta} \frac{\delta^2 F[h]}{\delta h(\vec{x}) \delta h(\vec{y})} = \frac{1}{\beta} K^{-1}(\vec{x} - \vec{y}), \quad (46)$$

$$= \frac{1}{V} \sum_{\vec{k}} \frac{e^{i\vec{k} \cdot (\vec{x} - \vec{y})}}{\bar{c}k^2 + \bar{\rho}(t)} \quad (47)$$

$$\xrightarrow{V \rightarrow \infty} \int \frac{d\vec{k}}{(2\pi)^d} \frac{e^{i\vec{k} \cdot (\vec{x} - \vec{y})}}{\bar{c}k^2 + \bar{\rho}(t)}. \quad (48)$$

Gaussian Model: critical exponents

Critical exponents follow from the previously obtained expressions:

- correlation length,

$$\xi \propto |T - T_0|^{-\nu_G}, \quad \nu_G = \nu_{MF} = \frac{1}{2}; \quad (49)$$

- correlation function for $r \ll \xi \rightarrow +\infty$,

$$G(r, T_0) \sim \frac{1}{r^{d-2+\eta_G}}, \quad \eta_G = \eta_{MF} = 0; \quad (50)$$

- isothermal susceptibility χ_T ,

$$\chi_T = \beta \int d\vec{r} G(\vec{r}), \quad (51)$$

$$= \int d\vec{r} \frac{1}{V} \sum_{\vec{k}} \frac{1}{\bar{c}k^2 + \bar{\rho}(t)} e^{i\vec{k} \cdot \vec{r}} = \frac{1}{\bar{c}} \sum_{\vec{k}} \frac{\delta_{\vec{k},0}}{k^2 + \xi^{-2}} \quad (52)$$

$$= \frac{\xi^2}{\bar{c}} \propto |T - T_0|^{-\gamma_G}, \quad \gamma_G = \gamma_{MF} = 1. \quad (53)$$

Gaussian Model: critical exponents

- specific heat c_V

$$c_V = -\frac{T}{V} \frac{\partial^2 F}{\partial T^2} \quad (54)$$

$$= \frac{k_B T}{V} \frac{\partial^2}{\partial T^2} (T \log Z_G[0]) \quad (55)$$

$$= \frac{k_B T}{V} \frac{\partial}{\partial T} \left(\log Z_G[0] + T \frac{\partial}{\partial T} \log Z_G[0] \right) \quad (56)$$

$$= k_B \left(T^2 \frac{\partial^2}{\partial T^2} + 2T \frac{\partial}{\partial T} \right) \left[\frac{1}{2V} \sum_{\vec{k}} \log \frac{2\pi k_B T V}{\bar{c} k^2 + \bar{\rho}(t)} \right] \quad (57)$$

$$= \frac{k_B}{2V} \sum_{\vec{k}} \left[1 - \frac{2T\bar{r}_0}{\bar{c} k^2 + \bar{\rho}(t)} + \frac{T^2 \bar{r}_0^2}{(\bar{c} k^2 + \bar{\rho}(t))^2} \right] \quad (58)$$

$$= c_V^{(1)} + c_V^{(2)} + c_V^{(3)} \quad (59)$$

where $\bar{r}_0 = d\bar{\rho}/dT$ is independent of T .

Gaussian Model: critical exponents

The term $c_V^{(1)}$ corresponds to Equipartition Theorem, i.e. $\frac{1}{2}k_B$ for each degree of freedom.

The term $c_V^{(2)}$ is proportional to

$$\int_0^\Lambda dk \frac{k^{d-1}}{\bar{c}k^2 + \bar{\rho}(t)} = \bar{\rho}(t)^{\frac{d-2}{2}} (\bar{c})^{-d/2} \int_0^{\bar{c}^{1/2}\Lambda/\bar{\rho}(t)^{1/2}} d\kappa \frac{\kappa^{d-1}}{\kappa^2 + 1} \quad (60)$$

$$\approx \begin{cases} \frac{\Lambda^{d-2}}{\bar{c}(d-2)} & d > 2, \Lambda < +\infty \\ \frac{1}{\bar{c}} \log \left(\left(\frac{\bar{c}}{\bar{\rho}(t)} \right)^{\frac{1}{2}} \Lambda \right) & d = 2, \Lambda < +\infty \\ \frac{1}{\bar{\rho}(t)^{1/2}} \frac{\pi}{2\bar{c}^{1/2}} & d = 1, \Lambda \rightarrow +\infty \end{cases} \quad (61)$$

Gaussian Model: critical exponents

The term $c_V^{(3)}$ is proportional to

$$\int_0^\Lambda dk \frac{k^{d-1}}{(\bar{c}k^2 + \bar{\rho}(t))^2} = \bar{\rho}(t)^{\frac{d-4}{2}} (\bar{c})^{-d/2} \int_0^{\bar{c}^{1/2}\Lambda/\bar{\rho}(t)^{1/2}} d\kappa \frac{\kappa^{d-1}}{(\kappa^2 + 1)^2} \quad (62)$$

$$\approx \begin{cases} \frac{\Lambda^{d-4}}{\bar{c}^2(d-4)} & d > 4, \Lambda < +\infty \\ \frac{\bar{\rho}(t)^{\frac{d-4}{2}}}{\bar{c}^{d/2}} \log \left(\left(\frac{\bar{c}}{\bar{\rho}(t)} \right)^{\frac{1}{2}} \Lambda \right) & d = 4, \Lambda < +\infty \\ \text{const.} \times (\bar{c})^{-d/2} \bar{\rho}(t)^{\frac{d-4}{2}} & d < 4, \Lambda \rightarrow +\infty. \end{cases} \quad (63)$$

For $d < 4$, c_V is IR divergent at criticality, with *non-MF behavior*

$$c_V \sim \rho(t)^{\frac{d-4}{2}} \propto |T - T_0|^{-\alpha_G}, \quad \alpha_G = 2 - \frac{d}{2} \neq \alpha_{MF} = 0 \quad (64)$$

Unfortunately, this is incorrect for more general models.